# The Relationship Between Obesity and Health, Lifestyle, and Economic Factors

System Analysis & Database Design Documentation

## 1. System Analysis & Design

### 1.1 Problem Statement & Objectives

**Problem Statement:** Raw global health data (WHO, NCD-RisC) and economic metrics are unstructured and fragmented, making comparative cross-country analysis difficult.

**Project Goals: Incorporate Health & Economic Indicators:** Build a structured SQL Server relational data warehouse (Health_Analytics) to integrate epidemiological health metrics (obesity prevalence, diabetes rates, and physical inactivity) alongside macroeconomic indicators (GDP per capita and current health expenditure per capita).

**Geographical & Temporal Scope Analysis:** Focus analytical tracking on specific cross-country metrics for Egypt, Japan, and the United States spanning a 23-year temporal timeline (2000–2022).

**Automated ETL Pipeline & Star Schema Modeling:** Implement robust data cleaning and pipeline workflows using Python (Pandas) to transform raw datasets into a normalized Star Schema architecture (DimCountry, DimYear, DimSex, and Fact tables) to ensure data integrity through composite primary and foreign key constraints.

**Cross-Factor & Correlation Analysis:** Analyze and identify statistical relationships, trends, and impact patterns between obesity prevalence and other health indicators (diabetes, physical inactivity) as well as macroeconomic factors (GDP, health spending).

**Advanced Business Intelligence Visualization:** Establish an enterprise analytical reporting layer utilizing modular database views connected to interactive BI dashboards (Power BI / Tableau) to uncover and present multi-factor correlations clearly.

### 1.2 System Actors & Interactions

- **Data Analyst:** Cleans raw data via Python, audits nulls/duplicates, and exports normalized CSVs.
- **DBA:** Creates database schemas, manages keys/constraints, and builds analytical views.
- **BI Analyst:** Consumes analytical views via Power BI / Tableau dashboards for decision making.

### 1.3 Requirements & Architecture

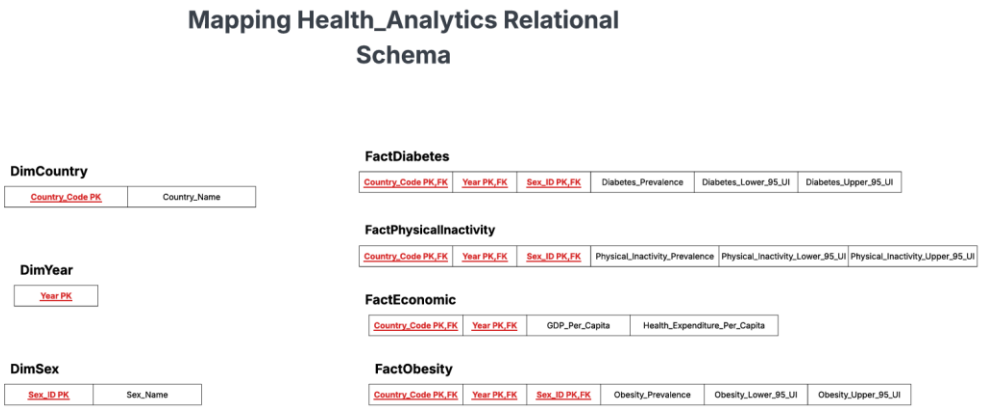
- **Functional Requirements:** ETL automation, strict referential integrity, and analytical view creation.
- **Non-Functional Requirements:** Sub-second query performance and ACID compliance.

**Architecture Style:** Star Schema design in MS SQL Server separating reference data into dimensions and metrics into fact tables.

# 2. Database Design & Data Modeling

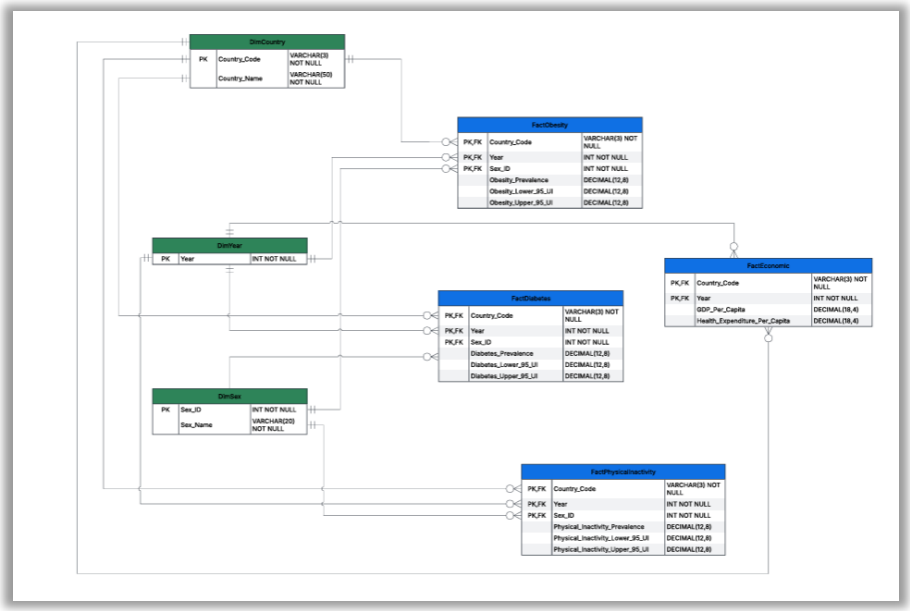
## 2.1 Entity-Relationship (ER) Model

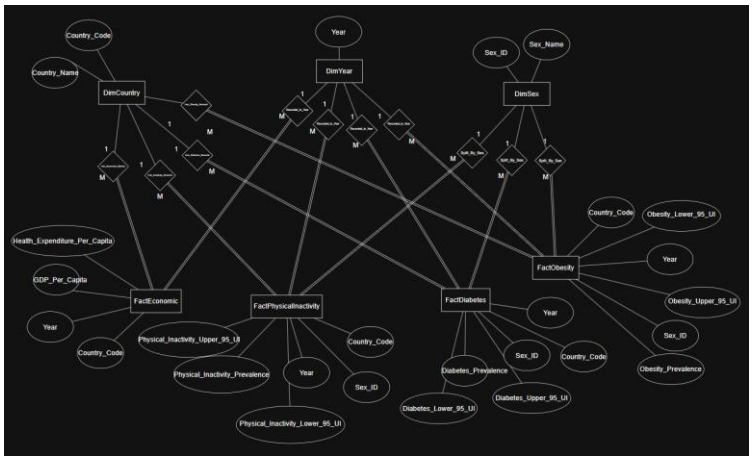
**Overview:** A Star Schema featuring 3 Dimension Tables (DimCountry, DimYear, DimSex) linked via 1:N relationships to 4 Fact Tables (FactObesity, FactDiabetes, FactPhysicalInactivity, FactEconomic).




## 2.2 Schema Specifications

| Table Name | Primary Key | Key Attributes |
|---|---|---|
| **DimCountry** | Country_Code | Country_Name |
| **DimYear** | Year | Years (2000–2022) |
| **DimSex** | Sex_ID | Sex_Name (Men, Women, Both sexes) |
| **Fact Tables** | Country_Code + Year (+ Sex_ID) | Prevalence / Economic metrics |

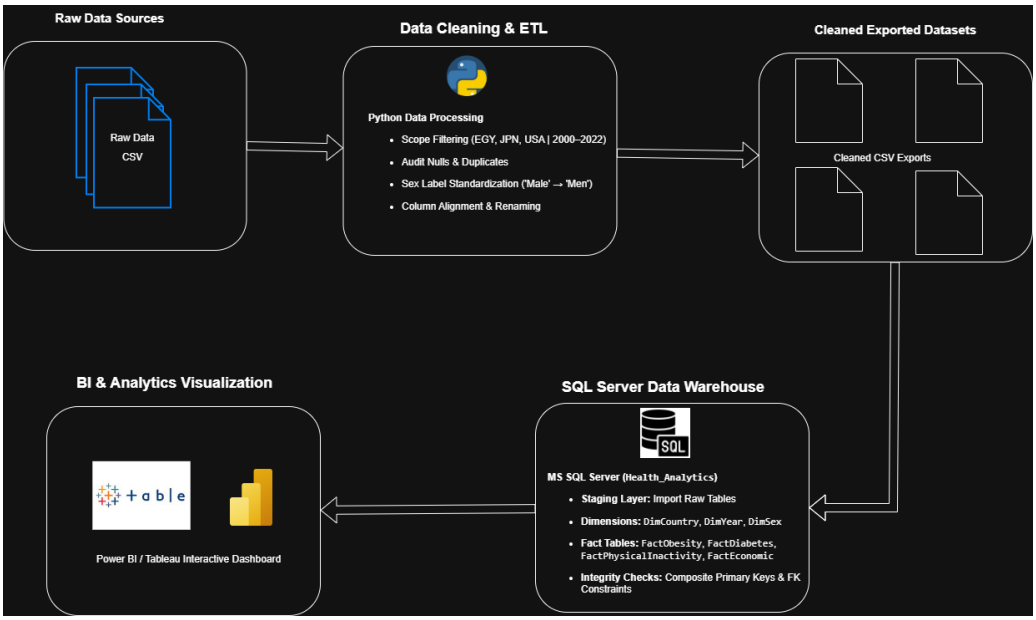





## 3. Data Flow & System Behavior

**Data Flow & Pipeline Architecture (DFD):**

1. **Raw Data Sources:** The process begins with raw epidemiological and economic data provided in CSV format.
2. **Data Cleaning & ETL:** Data is processed using Python (Pandas) for data cleaning and transformation. This includes scope filtering (Egypt, Japan, and the USA from 2000 to 2022), auditing null values and duplicates, standardizing sex labels, and column alignment.
3. **Cleaned Exported Datasets:** The cleaned datasets are exported into structured CSV files ready for loading.
4. **SQL Server Data Warehouse:** Data is loaded into the MS SQL Server database named Health_Analytics. This layer consists of a Staging Layer, Dimension tables (DimCountry, DimYear, DimSex), Fact tables (FactObesity, FactDiabetes, FactPhysicalInactivity, FactEconomic), and data integrity validation via composite primary keys and foreign key constraints.
5. **BI & Analytics Visualization:** In the final stage, the data is connected to create interactive dashboards and analytical reports using Business Intelligence tools such as Power BI and Tableau.




## 5. Technical Architecture & Validation

- **Tech Stack:** Python 3.14 (Pandas), MS SQL Server 17.0 (T-SQL), Power BI / Tableau.
- **Validation Plan:** Row count audits (138 health rows, 69 economic rows), zero-duplicate checks via HAVING, DBCC CHECKCONSTRAINTS.